NAME: Arya Raut
Roll no.: 2401093
Batch: T21

EXPERIMENT 1

Aim: Study and Implementation of Basic Git and GitHub Operations.

This Experiment maps to LO2.
LO2: To understand and perform the basic version control operations using Git and GitHub, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory:
Git is a distributed version control system (VCS) used to track changes in source code during software development. It allows developers to manage project versions, collaborate efficiently, maintain a history of changes, and restore previous versions when needed. Git stores project history locally, making it fast, reliable, and suitable for both individual and team development.
GitHub is a cloud-based hosting platform for Git repositories. It provides a central location where developers can store, share, and collaborate on projects using Git. GitHub offers additional features such as pull requests, issue tracking, project management, code reviews, and continuous integration.
difference between git and git hub:

Git	GitHub
Version control system	Cloud hosting platform
Installed locally	Accessible through the internet
Tracks source code changes	Stores Git repositories online
Can work without internet	Requires internet for remote operations
Open-source software	Web service built around Git

No.	Command	Purpose
1	`git --version`	Checks the installed Git version.
2	`git config --global user.name "Your Name"`	Sets the Git username globally.

NAME: Arya Raut
Roll no.: 2401093
Batch: T21

3	`git config --global user.email "email@example.com"`	Sets the Git email globally.
4	`git config --list`	Displays Git configuration details.
5	`mkdir gitproject`	Creates a new project folder.
6	`cd gitproject`	Navigates into the project directory.
7	`git init`	Initializes a new Git repository.
8	`echo "Hello Git" > file.txt`	Creates a new file with content.
9	`type file.txt` *(Windows)*	Displays the contents of the file.
10	`git status`	Shows the current repository status.
11	`git add file.txt`	Stages a specific file for commit.
12	`git add .`	Stages all modified files.
13	`git commit -m "Initial commit"`	Creates a commit with a message.
14	`git log`	Displays the complete commit history.
15	`git log --oneline`	Displays the commit history in short form.
16	`echo "Git Version Control" >> file.txt`	Appends content to an existing file.
17	`git diff`	Shows differences between working and committed files.
18	`git restore file.txt`	Restores an uncommitted file to its previous state.
19	`git reset HEAD~1`	Removes the latest commit while keeping changes.
20	`git branch feature`	Creates a new branch named **feature**.
21	`git branch`	Lists all available branches.

NAME: Arya Raut
Roll no.: 2401093
Batch: T21

22	`git checkout feature`	Switches to the **feature** branch.
23	`git switch feature`	Switches to the **feature** branch using the modern command.
24	`echo "Feature Branch" > feature.txt`	Creates a file in the feature branch.
25	`git checkout main`	Switches back to the **main** branch.
26	`git switch main`	Alternative command to switch to **main**.
27	`git merge feature`	Merges the **feature** branch into **main**.
28	`git branch -d feature`	Deletes the **feature** branch after merging.
29	`git remote add origin <repository-url>`	Connects the local repository to a GitHub repository.
30	`git remote -v`	Displays the configured remote repositories.
31	`git push -u origin main`	Pushes the local repository to GitHub and sets the upstream branch.
32	`git pull origin main`	Downloads and merges changes from GitHub.
33	`git clone <repository-url>`	Copies a remote repository to the local machine.

Implementation:

```
C:\Users\AI&DS>git --version
git version 2.55.0.windows.3
```

```
PS C:\Users\AI&DS> git config --global user.name "arya tsec"
PS C:\Users\AI&DS> get config --global user.email "aryatsec093@gmail.com"
```

NAME: Arya Raut
Roll no.: 2401093
Batch: T21

```
PS C:\Users\AI&DS> git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=aryatsec093@gmail.com
user.name=arya tsec
PS C:\Users\AI&DS>
```

```
PS C:\Users\AI&DS> mkdir git-project


    Directory: C:\Users\AI&DS


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        23-07-2026     11:46                git-project


PS C:\Users\AI&DS> cd git-project
PS C:\Users\AI&DS\git-project> git init
Initialized empty Git repository in C:/Users/AI&DS/git-project/.git/
PS C:\Users\AI&DS\git-project> echo "arya was here"> read.txt
PS C:\Users\AI&DS\git-project> cat read.txt
arya was here
```

NAME: Arya Raut
Roll no.: 2401093
Batch: T21

```
PS C:\Users\AI&DS\git-project> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        read.txt

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\AI&DS\git-project> git add read.txt
PS C:\Users\AI&DS\git-project> git commit -m "first commit"
[master (root-commit) 7d7f86b] first commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 read.txt
PS C:\Users\AI&DS\git-project> git log --oneline
7d7f86b (HEAD -> master) first commit
PS C:\Users\AI&DS\git-project> echo "[REDACTED]" >> read.txt
PS C:\Users\AI&DS\git-project> git diff
diff --git a/read.txt b/read.txt
index 301335d..cca8c65 100644
Binary files a/read.txt and b/read.txt differ
PS C:\Users\AI&DS\git-project> git restore read.txt
```

```
PS C:\Users\AI&DS\git-project> git branch feature
PS C:\Users\AI&DS\git-project> git branch
  feature
* master
PS C:\Users\AI&DS\git-project> git checkout feature
Switched to branch 'feature'
PS C:\Users\AI&DS\git-project> git switch
fatal: missing branch or commit argument
PS C:\Users\AI&DS\git-project> git switch feature
Already on 'feature'
```

NAME: Arya Raut
Roll no.: 2401093
Batch: T21

```
PS C:\Users\AI&DS\git-project> echo "arya was here">newbranch.txt
PS C:\Users\AI&DS\git-project> git checkout main
error: pathspec 'main' did not match any file(s) known to git
PS C:\Users\AI&DS\git-project> git checkout master
Switched to branch 'master'
PS C:\Users\AI&DS\git-project> git switch master
Already on 'master'
PS C:\Users\AI&DS\git-project> git merge feature
Already up to date.
PS C:\Users\AI&DS\git-project> git branch -d feature
Deleted branch feature (was 7d7f86b).
PS C:\Users\AI&DS\git-project> git remote -v
PS C:\Users\AI&DS\git-project> git push -u origin master
```

```
PS C:\Users\AI&DS\git-project> git clone https://github.com/aryatsec093/git-practice
Cloning into 'git-practice'...
warning: You appear to have cloned an empty repository.
```